

Audio Equalizer and Analyzer

Anish Toshniwal (2024102009)

Anushka Saini (2024102052)

May 6, 2026

Abstract

This report details the design, simulation, and hardware implementation of a 5-band audio equalizer. The system utilizes multiple-feedback (MFB) Butterworth bandpass filters to isolate distinct audio frequency bands ranging from 20 Hz to 20 kHz. The filtered signals are processed through variable gain controls, peak envelope detectors, a summing amplifier, and a Class AB power amplifier. The system also includes an audio spectrum analyzer matrix. The report evaluates the theoretical calculations against simulated and measured results, providing a comprehensive analysis of component tolerances, parasitic effects, and design limitations.

Contents

1	Introduction	1
2	System Overview	1
2.1	Block Diagram	1
2.2	Circuit Diagram	2
3	Filter Stage Design: 5 Bandpass Filters	2
3.1	Frequency Band Walkthrough	2
3.2	Filter Topology Analysis	3
3.2.1	Topology Comparison: MFB vs. Sallen-Key	3
3.2.2	Butterworth Response and Filter Selection	3
3.3	Calculations and Implementation	4
4	Signal Processing and Output Stages	4
4.1	Variable Gain and Summing Amplifier	4
4.1.1	Variable Gain Control	4
4.1.2	Signal Recombination via Summing Amplifier	4
4.1.3	Summing Amplifier Implementation	5
4.2	Envelope Detector	6
4.2.1	Implementation and Calculations	6
4.2.2	Design Justification	7
4.3	Audio Analyzer Matrix	8
4.3.1	Architecture and Calculation	8
4.3.2	Principle of Operation	9
4.4	Class AB Power Amplifier	10
4.4.1	Transistor Selection	10
4.4.2	Biasing Network and Derivation	10
4.4.3	Expected Output Power	11
5	Simulation and Measured Results	12
5.1	LTSpice Simulations	12
5.1.1	Frequency Responese Analysis for whole circuit	12
5.1.2	Outputs for the a few bands a specific frequencies with gain = 0dB for all gain stages	13

5.2	Hardware Circuit	15
5.2.1	Bode Plots:	16
5.2.2	Inputs	19
5.2.3	Analyzer	19
6	Measurements and Experimental Observations	20
7	Analysis and Discussion	20
7.1	Cross-Band Interference	20
7.2	Complex Fourth Order Bandpass Filter Design	21
7.3	Phase Response Mismatch	21
7.4	Maintaining Audio Quality	21
7.5	Parasitic Effects	21
7.6	Suggestions for Improvement	21
8	Conclusion	22

1 Introduction

The objective of this project is to develop a 5-band audio equalizer capable of enhancing and customizing audio quality by independently adjusting specific frequency ranges according to user preference. Equalizers are fundamental components in audio engineering, allowing for the compensation of room acoustics, speaker deficiencies, or personal aesthetic preferences. This project demonstrates the practical application of active analog filtering, signal summation, and power amplification.

2 System Overview

2.1 Block Diagram

The architecture of the 5-band audio equalizer operates through a parallel processing topology. The implementation involves splitting the input audio signal into five distinct frequency bands using fourth-order multi-feedback (MFB) Butterworth bandpass filters. The gain of each band is independently controlled using potentiometers to adjust different frequency ranges according to user preference. The filtered signals are then combined using a summing amplifier to generate the modified audio output, which is reproduced through a speaker. Furthermore, the system includes a frequency response display for analyzing the effect of gain variation across different bands, demonstrating the practical implementation of audio signal processing, active filters, and operational amplifier-based control circuits.

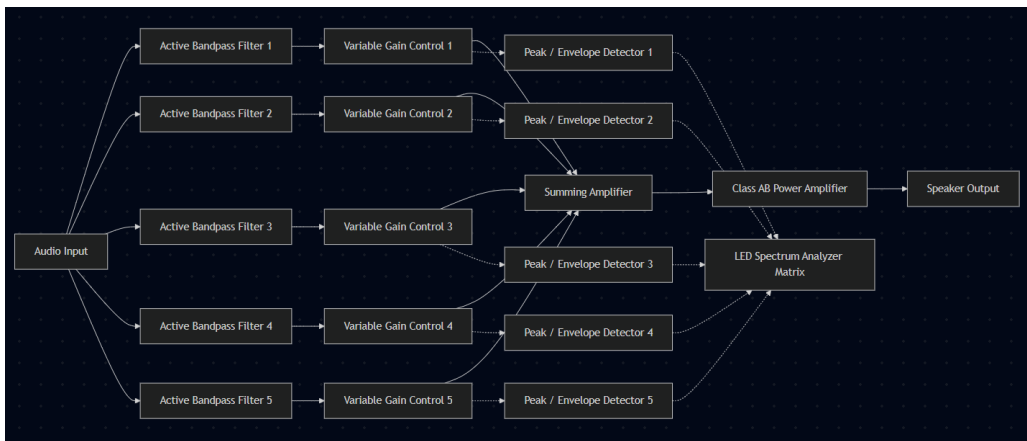


Figure 1: System block diagram illustrating signal flow from audio input to speaker and audio analyzer output.

2.2 Circuit Diagram

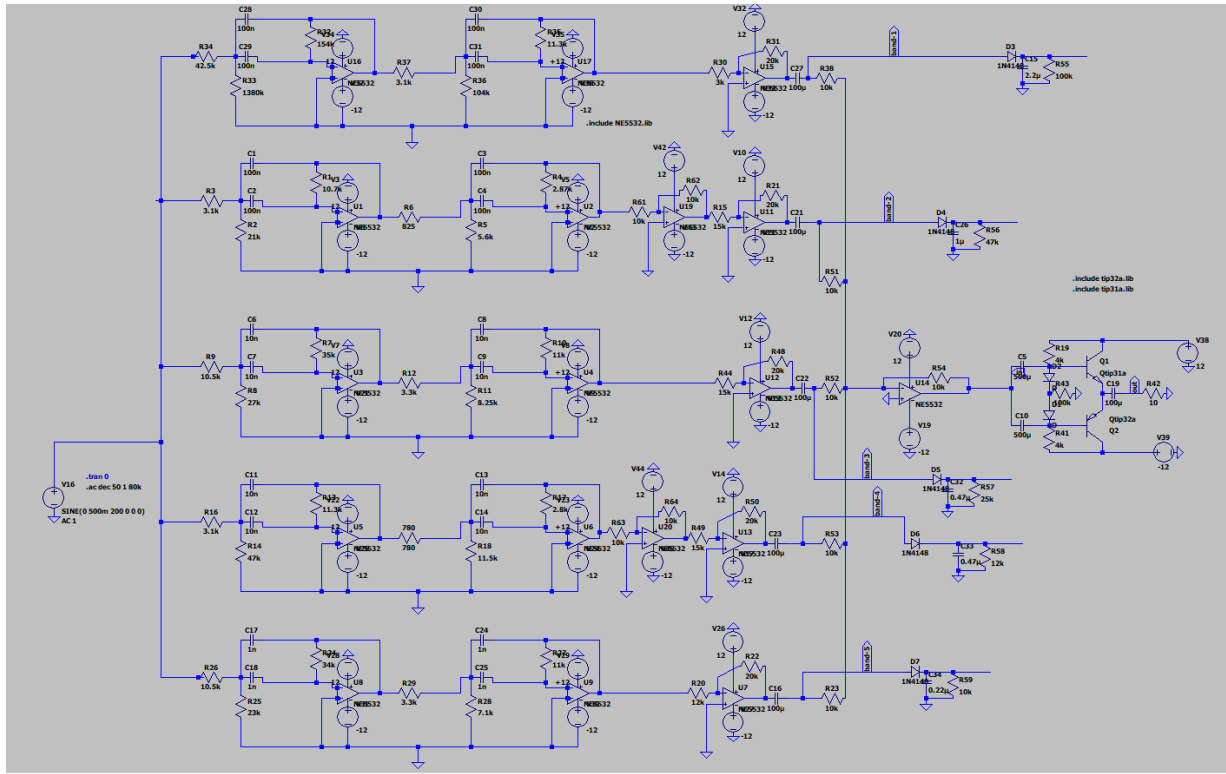


Figure 2: Complete schematic diagram of the 5-band audio equalizer.

3 Filter Stage Design: 5 Bandpass Filters

The core of the frequency separation relies on active bandpass filters. The audio spectrum (20 Hz to 20 kHz) is divided into five adjacent bands.

3.1 Frequency Band Walkthrough

The system targets the following specific frequency ranges, ensuring complete coverage of the audible spectrum:

- **Band 1 (Sub-Bass/Bass):** 20 Hz – 350 Hz
- **Band 2 (Low-Midrange):** 350 Hz – 1 kHz
- **Band 3 (Midrange):** 1 kHz – 3 kHz
- **Band 4 (High-Midrange):** 3 kHz – 10 kHz
- **Band 5 (Treble/Air):** 10 kHz – 20 kHz

3.2 Filter Topology Analysis

To achieve the desired frequency isolation, an active filter topology must be selected.

3.2.1 Topology Comparison: MFB vs. Sallen-Key

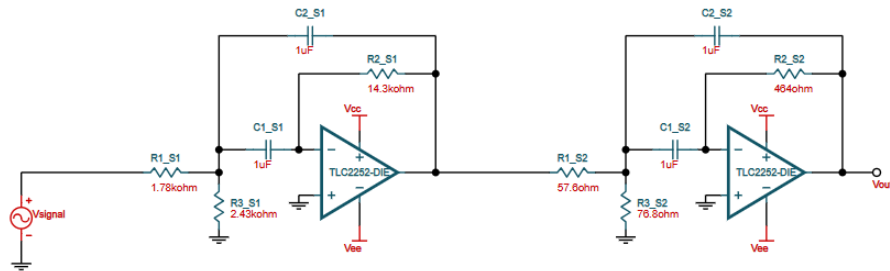
When implementing active bandpass filters, it is crucial to select a topology that meets the demands of audio equalization. The following table compares the characteristics of Multiple Feedback (MFB) and Sallen-Key topologies to justify our design choice:

Table 1: Comparison between Multiple Feedback (MFB) and Sallen-Key Topologies

Multiple Feedback (MFB)	Sallen-Key
Higher Q-factor and sharper band separation	Lower selectivity for bandpass filters
Better gain and Q control	Limited high-Q performance
More stable for narrow bandwidth filters	Can become unstable at high Q
Reduced overlap between adjacent bands	More cross-band interference
Better suited for bandpass equalizer applications	Mostly preferred for low-pass/high-pass filters

3.2.2 Butterworth Response and Filter Selection

The project uses fourth order multi-feedback (MFB) Butterworth bandpass filters to accurately separate the input audio signal into distinct frequency bands with sharp selectivity and minimal band overlap. The Butterworth response ensures a maximally flat and low-distortion output, preserving audio quality, while the MFB topology provides high stability, compact implementation, efficient gain control, and reliable performance, making it highly suitable for practical audio equalizer applications.



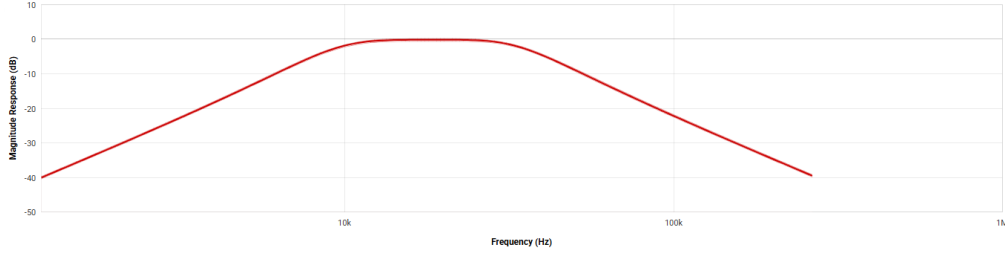


Figure 3: Schematic of the fourth-order MFB Butterworth bandpass filter and its corresponding magnitude response.

3.3 Calculations and Implementation

To adhere to the stringent 25-page limit of this report, the comprehensive step-by-step mathematical derivations and component calculations for all five MFB filter bands is added in the link below:

<https://drive.google.com/drive/folders/1HjKUTuw6CYNPMXkxBrci83NWthcEkepB>

4 Signal Processing and Output Stages

4.1 Variable Gain and Summing Amplifier

Following the active bandpass filter stage, the audio signal has been separated into five distinct frequency bands. To achieve the equalization effect, the user must be able to independently modify the amplitude (gain) of each specific frequency component before they are recombined.

4.1.1 Variable Gain Control

The output of each of the five bandpass filters, denoted as V_{BP1} through V_{BP5} , is routed through a linear potentiometer configured as a voltage divider. This allows the user to apply an attenuation factor, α_i (where $0 \leq \alpha_i \leq 1$), to each band. The adjusted voltage for each band, $V_{in,i}$, entering the summation stage is given by:

$$V_{in,i} = \alpha_i \cdot V_{BPi} \quad \text{for } i \in \{1, 2, 3, 4, 5\} \quad (1)$$

4.1.2 Signal Recombination via Summing Amplifier

To reconstruct the full audio spectrum, these individually adjusted frequency bands must be added back together. This is achieved using an inverting summing amplifier configuration.

The theoretical output voltage V_{out} of an inverting summing amplifier with a feedback resistor R_f and individual input resistors $R_{in,1}$ through $R_{in,5}$ is defined by Kirchhoff's Current Law at the inverting

node:

$$V_{out} = -R_f \left(\frac{V_{in,1}}{R_{in,1}} + \frac{V_{in,2}}{R_{in,2}} + \frac{V_{in,3}}{R_{in,3}} + \frac{V_{in,4}}{R_{in,4}} + \frac{V_{in,5}}{R_{in,5}} \right) \quad (2)$$

To ensure that the summing amplifier recombines the signals with a baseline unity gain—meaning the potentiometers serve as the sole amplitude control—all input resistors are chosen to be equal to the feedback resistor ($R_{in,i} = R_f$). This simplifies the derivation to:

$$V_{out} = -\frac{R_f}{R_{in}} \sum_{i=1}^5 V_{in,i} = -\sum_{i=1}^5 \alpha_i V_{BPI} \quad (3)$$

4.1.3 Summing Amplifier Implementation

Based on the system design, the summing amplifier is implemented using an NE5532 low-noise audio operational amplifier, powered by a dual supply of $\pm 12V$ to provide ample headroom and prevent signal clipping.

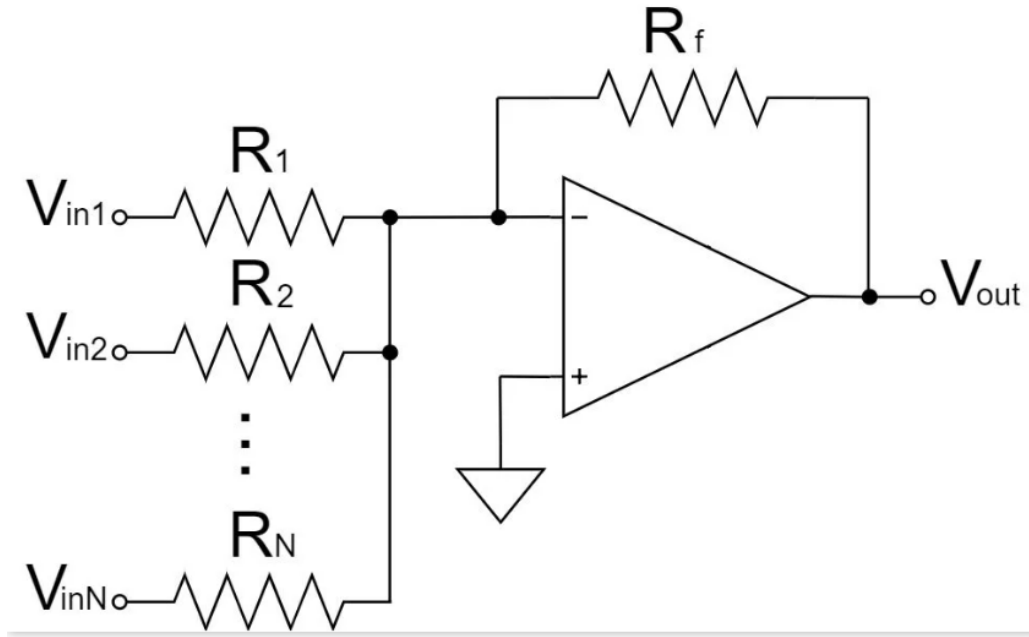


Figure 4: Schematic of the NE5532 inverting summing amplifier

In our circuit, the feedback resistor (R_f) is set to $10k\Omega$. To maintain unity gain during summation, the five input resistors connecting the potentiometers to the inverting terminal of the NE5532 are also set to $10k\Omega$. Substituting these practical values into our derived equation yields:

$$V_{out} = -\frac{10k\Omega}{10k\Omega} (V_{in,1} + V_{in,2} + V_{in,3} + V_{in,4} + V_{in,5}) \quad (4)$$

$$V_{out} = -(V_{in,1} + V_{in,2} + V_{in,3} + V_{in,4} + V_{in,5}) \quad (5)$$

This confirms that the circuit successfully reconstructs the equalized audio signal. Note that the output is inverted (180° phase shift) relative to the inputs.

4.2 Envelope Detector

To provide visual feedback, the output of each bandpass filter is tapped by a peak/envelope detector. This circuit converts the AC audio signal into a varying DC voltage proportional to the signal's amplitude within that specific band.

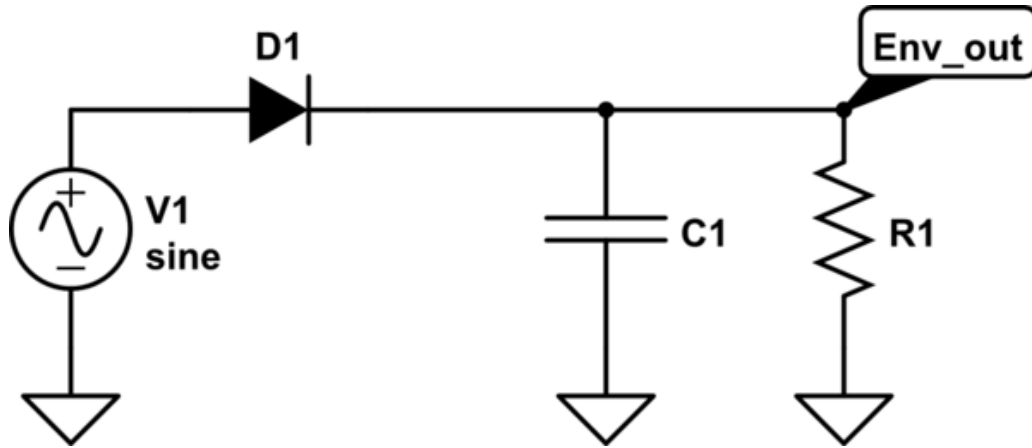


Figure 5: Envelope Detector

4.2.1 Implementation and Calculations

The implementation utilizes 1N4148 high-speed switching diodes for all envelope as a half-wave rectifier to focus only on the positive half cycle. This diode introduces a forward voltage drop ($V_f \approx 0.7V$). Consequently, the audio signal must exceed $0.7V$ to charge the capacitor. In the context of a visualizer, this acts beneficially as a rudimentary "noise gate," preventing low-level background hiss from triggering the LEDs.

The dynamic response of the visualizer—how quickly the LEDs light up and fade out—is governed by the RC time constant (τ) of the capacitor-resistor network. The time constant defines the time required for the capacitor to discharge to approximately 36.8% of its peak voltage, calculated using the formula:

$$\tau = R \times C \quad (6)$$

To ensure optimal tracking across the entire audio spectrum, the component values are scaled according to their respective frequency bands. The calculations for all five bands are as follows:

- **Band 1 (Sub-Bass/Bass: 20 Hz – 300 Hz):** Given $C_{15} = 2.2\mu\text{F}$ and $R_{55} = 100\text{k}\Omega$.

$$\tau_1 = 100 \times 10^3 \Omega \times 2.2 \times 10^{-6} \text{F} = 0.220 \text{s} = 220 \text{ms}$$

- **Band 2 (Low-Midrange: 300 Hz – 1 kHz):** Given $C_{26} = 1\mu\text{F}$ and $R_{56} = 47\text{k}\Omega$.

$$\tau_2 = 47 \times 10^3 \Omega \times 1 \times 10^{-6} \text{F} = 0.047 \text{s} = 47 \text{ms}$$

- **Band 3 (Midrange: 1 kHz – 3 kHz):** Given $C_{32} = 0.47\mu\text{F}$ and $R_{57} = 25\text{k}\Omega$.

$$\tau_3 = 25 \times 10^3 \Omega \times 0.47 \times 10^{-6} \text{F} = 0.01175 \text{s} = 11.75 \text{ms}$$

- **Band 4 (High-Midrange: 3 kHz – 10 kHz):** Given $C_{33} = 0.47\mu\text{F}$ and $R_{58} = 12\text{k}\Omega$.

$$\tau_4 = 12 \times 10^3 \Omega \times 0.47 \times 10^{-6} \text{F} = 0.00564 \text{s} = 5.64 \text{ms}$$

- **Band 5 (Treble/Air: 10 kHz – 20 kHz):** Given $C_{34} = 0.22\mu\text{F}$ and $R_{59} = 10\text{k}\Omega$.

$$\tau_5 = 10 \times 10^3 \Omega \times 0.22 \times 10^{-6} \text{F} = 0.0022 \text{s} = 2.2 \text{ms}$$

4.2.2 Design Justification

The calculated time constants intentionally decrease as the frequency band increases, which is highly optimal for an audio spectrum analyzer. Lower frequency signals (like Band 1 at 20 Hz) have longer wavelengths and slower cycles. If τ_1 were too short, the capacitor would discharge rapidly between these slow cycles, causing the bass LEDs to visibly flicker. Therefore, a longer time constant of 220 ms is used to smooth the low-frequency envelope.

Conversely, high-frequency signals (like Band 5 up to 20 kHz) contain fast, transient musical elements like cymbal crashes. By reducing the time constant to 2.2 ms for Band 5, the DC output is able to track the rapid rhythmic envelope of treble sounds responsively. This scaled approach strikes the ideal balance, ensuring smooth and accurate visual representation across the entire audio spectrum.

4.3 Audio Analyzer Matrix

To provide real-time visual feedback of the equalization, the system features a 5-band LED spectrum analyzer matrix that translates the DC envelope voltages into proportional amplitude displays.

4.3.1 Architecture and Calculation

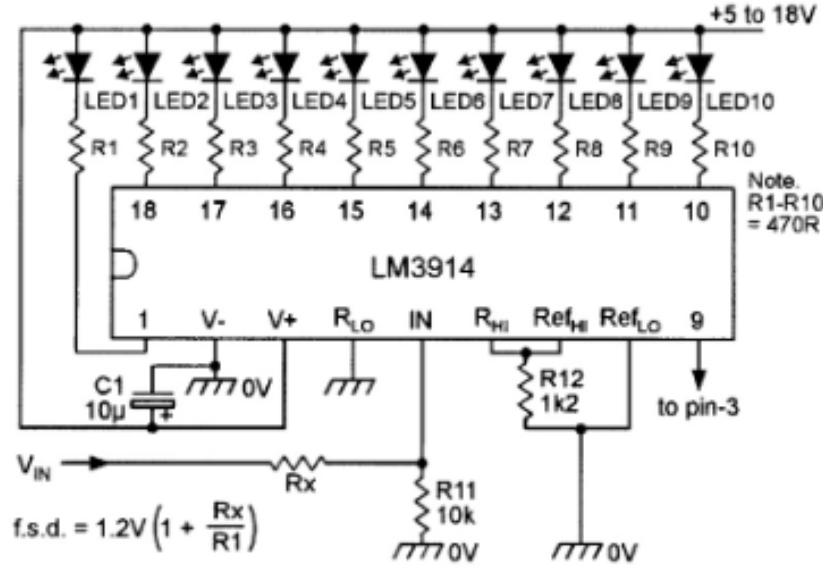


Figure 6: LM3914 in Bar display configuration

The matrix consists of five parallel strips of 9 LEDs (one strip per frequency band). These are driven by five **LM3914 Bar Display Driver ICs**.

The below formula (ignore the one in image) is used to set the value of DC voltage for the last LED to glow.

$$\text{f.s.d.} = 1.2\text{V} \left(1 + \frac{R_x}{R_{11}} \right) \quad (7)$$

Given an initial peak-to-peak voltage (V_{pp}) of 8.6 V (discussed further in section 6) from the filter stage, the theoretical maximum DC voltage (V_{max}) produced by the envelope detector is determined by halving the peak-to-peak value to find the AC amplitude, and then subtracting the forward voltage drop ($V_f \approx 0.7 \text{ V}$) of the rectifying diode:

$$V_{max} = \left(\frac{V_{pp}}{2} \right) - V_f \quad (8)$$

Substituting the known values into the equation yields the final maximum voltage stored by the capacitor:

$$V_{\max} = \left(\frac{8.6 \text{ V}}{2} \right) - 0.7 \text{ V}$$

$$V_{\max} = 4.3 \text{ V} - 0.7 \text{ V}$$

$$V_{\max} = 3.6 \text{ V}$$

Since the maximum envelope voltage is calculated as 3.6 V, we set the full-scale deflection (f.s.d.) of the display driver to match this peak (f.s.d. = 3.6 V).

Using the provided reference voltage formula and a standard chosen value of $R_{11} = 10 \text{ k}\Omega$, we can calculate the required value for the feedback resistor (R_x) as follows:

$$\text{f.s.d.} = 1.2 \text{ V} \left(1 + \frac{R_x}{R_{11}} \right)$$

$$3.6 \text{ V} = 1.2 \text{ V} \left(1 + \frac{R_x}{10 \text{ k}\Omega} \right)$$

$$\frac{3.6}{1.2} = 1 + \frac{R_x}{10 \text{ k}\Omega}$$

$$3 = 1 + \frac{R_x}{10 \text{ k}\Omega}$$

$$2 = \frac{R_x}{10 \text{ k}\Omega}$$

$$R_x = 2 \times 10 \text{ k}\Omega$$

$$R_x = 20 \text{ k}\Omega$$

4.3.2 Principle of Operation

The operation for each frequency band follows a direct threshold logic:

1. **Input:** The smoothed DC voltage from the envelope detector is fed into the signal input (Pin 5) of the LM3914.
2. **Comparison:** Inside the IC, this voltage is compared against a precision resistor-divider network connected to a bank of ten operational amplifiers acting as comparators.
3. **Illumination:** As the DC voltage rises with louder audio signals, it sequentially exceeds these internal thresholds. Each triggered comparator sinks current, pulling its output low and illuminating the corresponding LED in proportion to the signal's peak amplitude.

4.4 Class AB Power Amplifier

The final recombined audio signal from the summing amplifier is a voltage-level signal that lacks the current-driving capability necessary to actuate a standard low-impedance loudspeaker. To resolve this, a Class AB push-pull power amplifier is implemented. This topology provides the high current required by the load while utilizing a biasing network to effectively eliminate the crossover distortion inherent in pure Class B amplifiers.

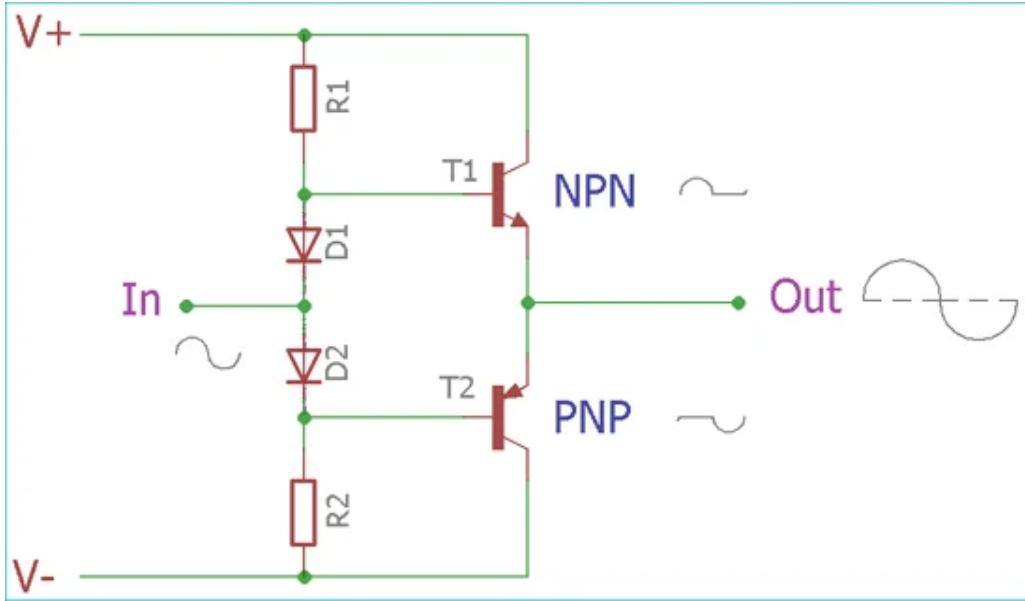


Figure 7: Schematic of the Class AB push-pull power amplifier featuring TIP31A/TIP32A complementary transistors and a diode biasing network.

4.4.1 Transistor Selection

As shown in Figure 7, the output stage utilizes a complementary pair of bipolar junction transistors (BJTs): the **TIP31A** (NPN, Q1) and the **TIP32A** (PNP, Q2). These are robust, medium-power transistors capable of handling continuous collector currents up to 3 A and significant power dissipation, making them highly suitable for driving a 10Ω audio load (R_{42}).

4.4.2 Biasing Network and Derivation

In a standard Class B push-pull amplifier, the transistors remain entirely off until the input audio signal exceeds their base-emitter turn-on voltage ($V_{BE} \approx 0.7V$). This dead zone creates severe crossover distortion at the zero-crossing of the AC signal.

To achieve Class AB operation, the transistors are slightly forward-biased using a constant voltage drop created by two series diodes. Because the diodes are manufactured from the same semiconductor

material as the transistors, their forward voltage drops closely track the V_{BE} requirements of the BJTs.

The biasing network is powered by the dual $\pm 12V$ supply (V_{CC} and V_{EE}) through identical $4k\Omega$ resistors (R_{19} and R_{41}). The required DC bias current (I_{bias}) flowing through this network to establish the voltage drop can be calculated using Kirchhoff's Voltage Law (KVL):

$$I_{bias} = \frac{V_{CC} - V_{EE} - 2V_D}{R_{19} + R_{41}} \quad (9)$$

Assuming a standard forward diode drop $V_D \approx 0.7V$:

$$I_{bias} = \frac{12V - (-12V) - 1.4V}{4000\Omega + 4000\Omega} = \frac{22.6V}{8000\Omega} \approx 2.825 \text{ mA} \quad (10)$$

This 2.825 mA quiescent current is sufficient to keep the diodes active. Consequently, the voltage difference between the bases of Q1 and Q2 is fixed at approximately 1.4V ($2 \times 0.7V$). Because $V_{B1} - V_{B2} = 1.4V$, both transistors are held exactly at the threshold of conduction ($V_{BE1} + |V_{BE2}| \approx 1.4V$). When the AC audio signal is coupled into the bases via the $500\mu F$ capacitors (C_5 and C_{10}), one transistor immediately conducts the positive half-cycle while the other smoothly takes over for the negative half-cycle without any dead zone.

4.4.3 Expected Output Power

The output signal is passed through a $100\mu F$ DC-blocking coupling capacitor to a 10Ω simulated speaker load. The theoretical maximum unclipped AC peak voltage (V_p) at the output is limited by the supply rails minus the saturation voltage of the transistors ($V_{CE(sat)} \approx 1.2V$ for TIP31/32).

Assuming a maximum practical peak output voltage swing of $V_p \approx 10V$, the expected Root Mean Square (RMS) voltage is:

$$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{10V}{1.414} \approx 7.07V \quad (11)$$

The expected maximum average audio power (P_{out}) delivered to the 10Ω load is calculated as:

$$P_{out} = \frac{V_{rms}^2}{R_L} = \frac{V_p^2}{2R_L} \quad (12)$$

$$P_{out} = \frac{(10V)^2}{2 \times 10\Omega} = \frac{100}{20} = 5 \text{ Watts} \quad (13)$$

This derivation demonstrates that the chosen configuration safely supplies 5 W of continuous power to the load, which is well within the thermal and current limits of the selected TIP31A/TIP32A transistors.

5 Simulation and Measured Results

5.1 LTSpice Simulations

5.1.1 Frequency Response Analysis for whole circuit

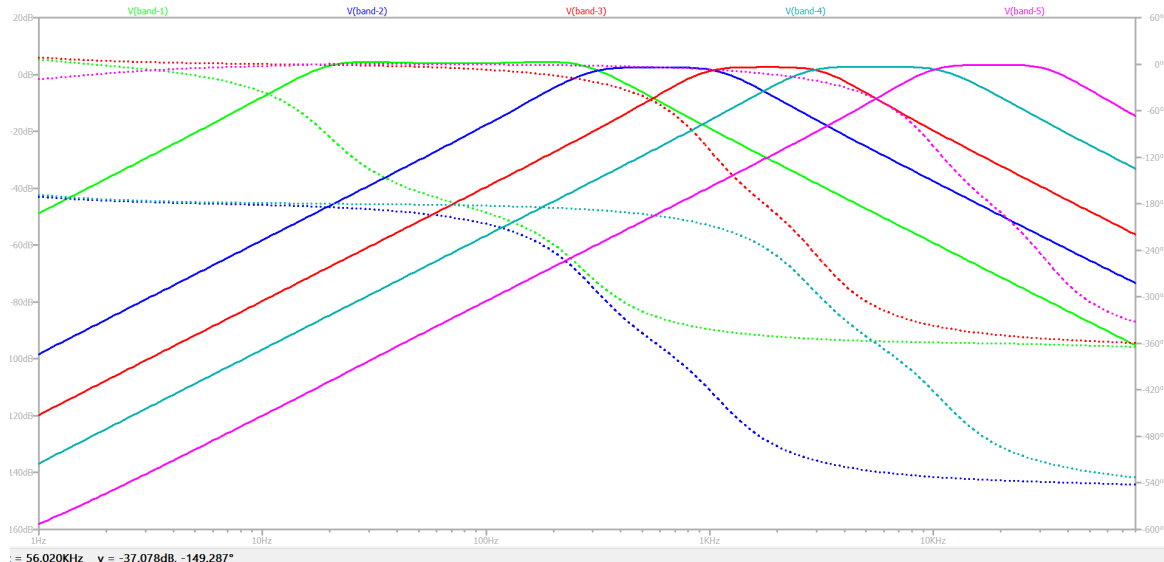


Figure 8: Bode plot showing response from 25 Hz to 20kHz in separated bands

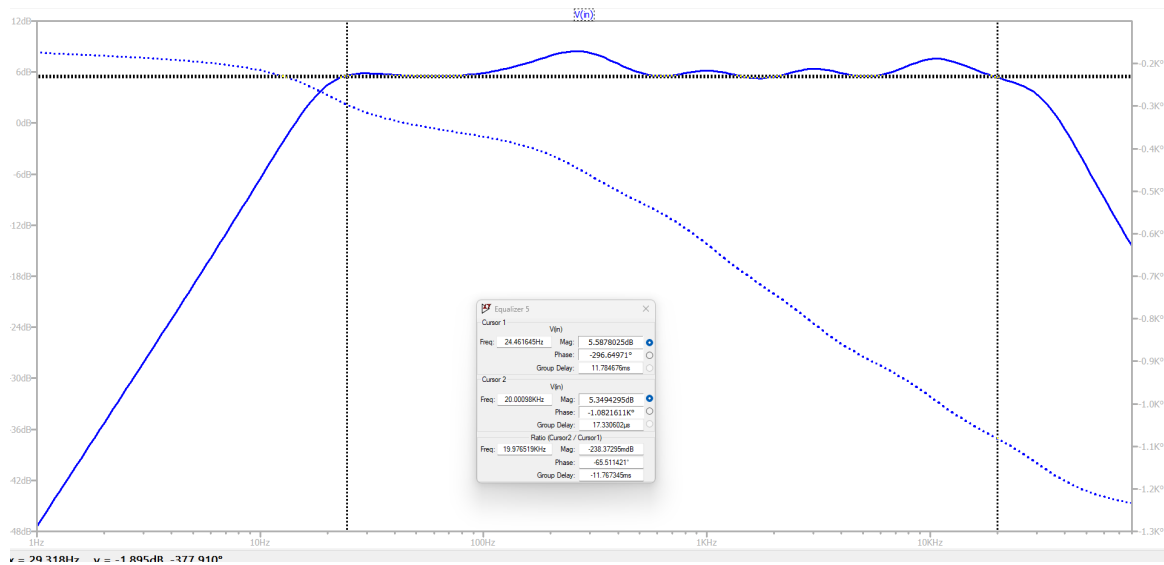


Figure 9: Bode plot showing response from 25 Hz to 20kHz by adding the separate band outputs

5.1.2 Outputs for the a few bands a specific frequencies with gain = 0dB for all gain stages

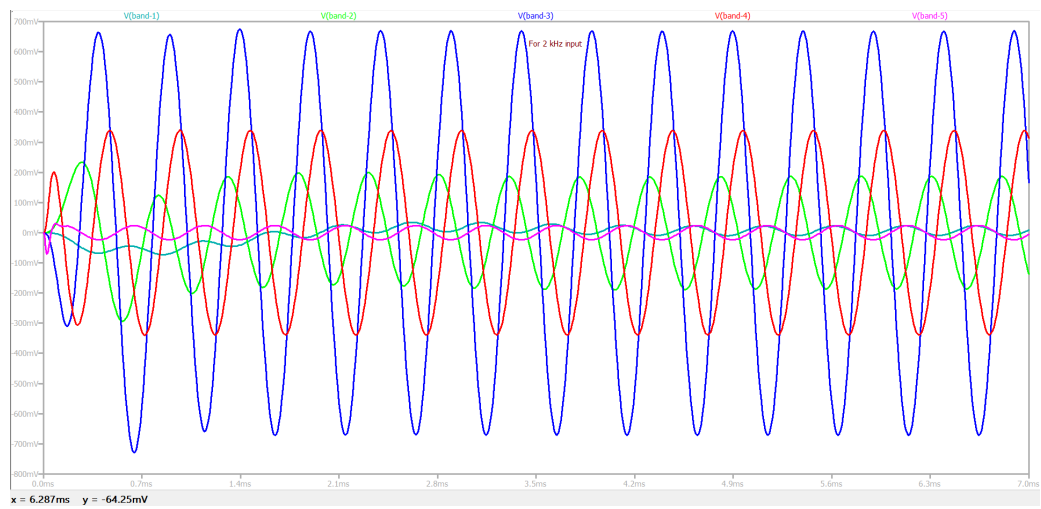


Figure 10: 2kHz bandwise output

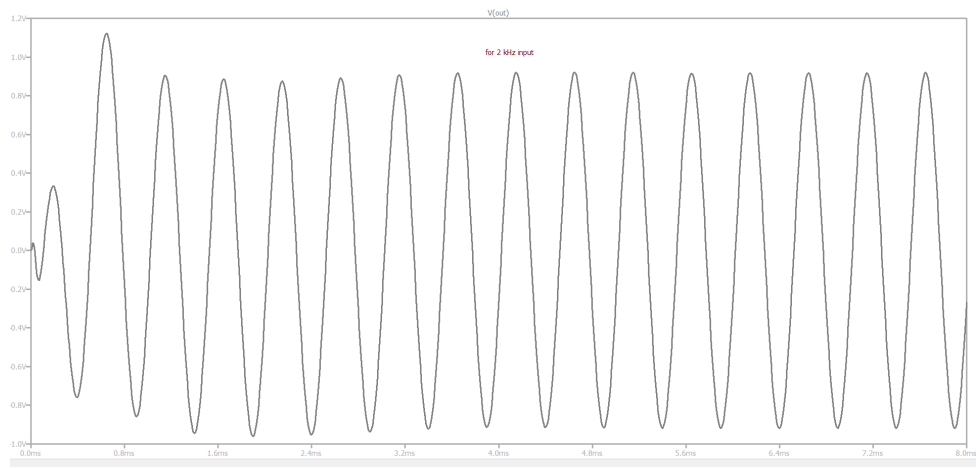


Figure 11: 2kHz output

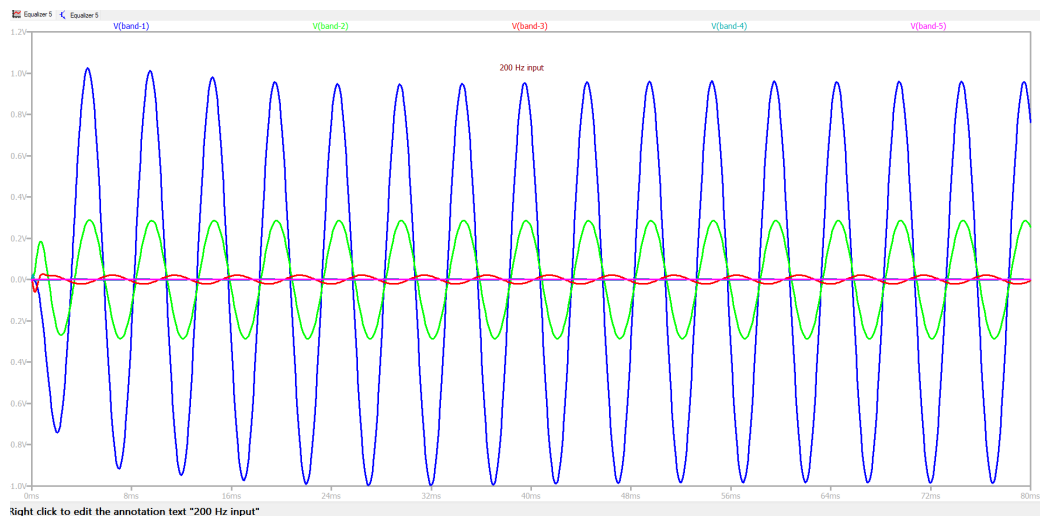


Figure 12: 200 Hz bandwise output

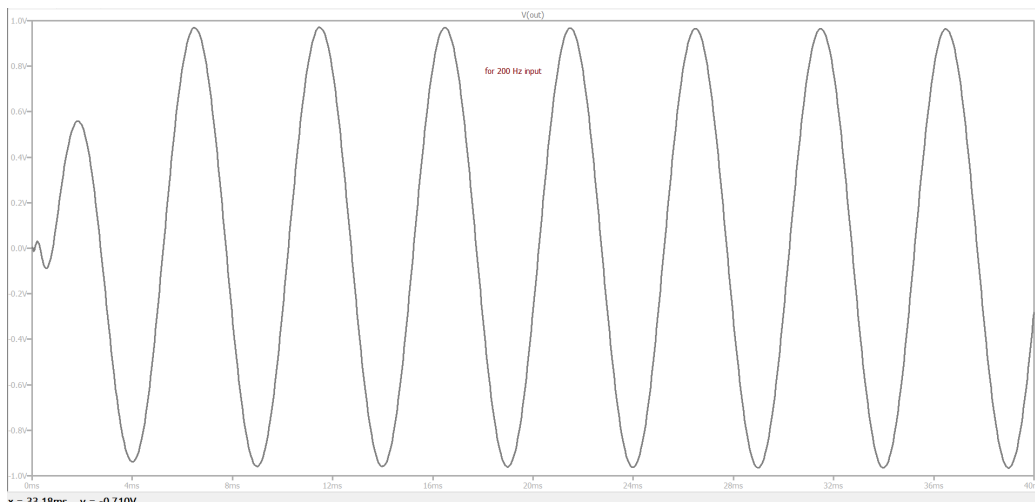


Figure 13: 200 Hz output

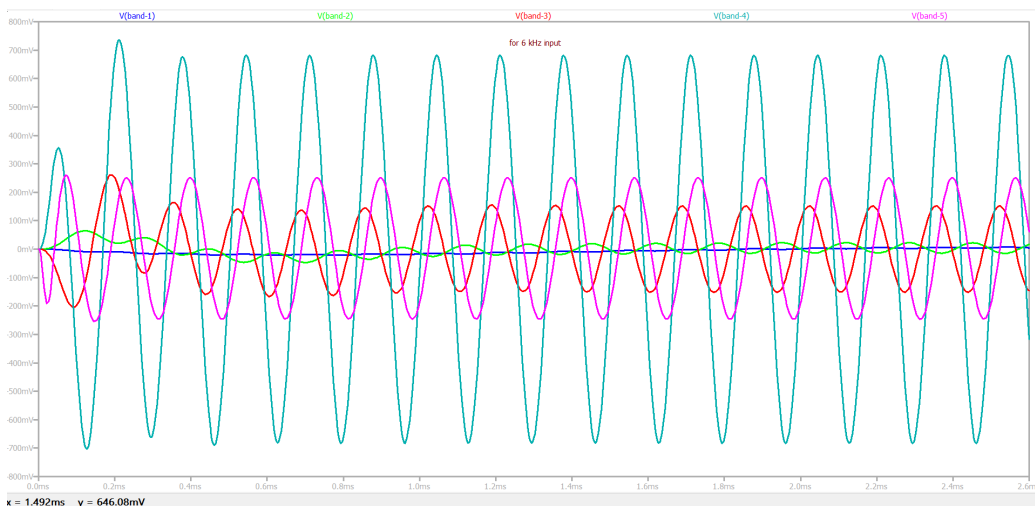


Figure 14: 6kHz bandwise output

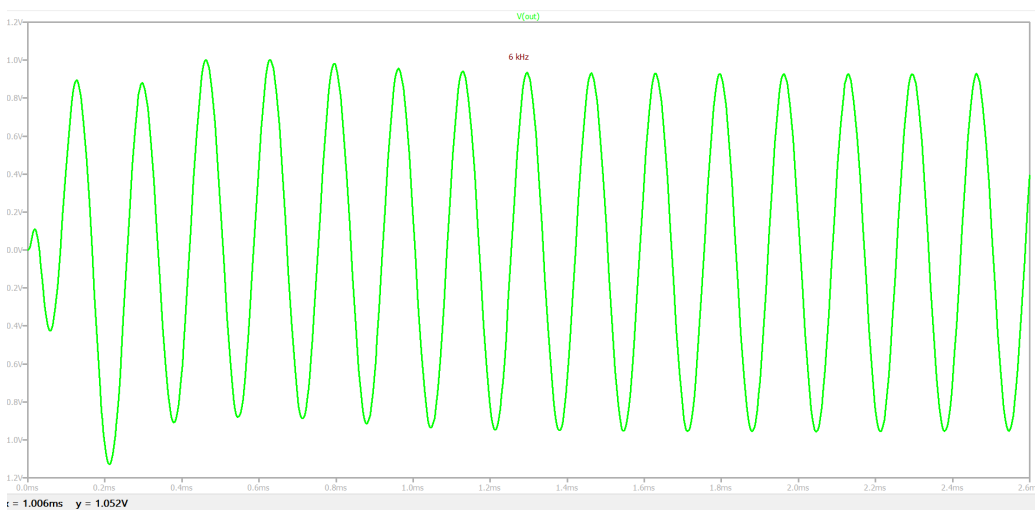


Figure 15: 6 kHz output

5.2 Hardware Circuit

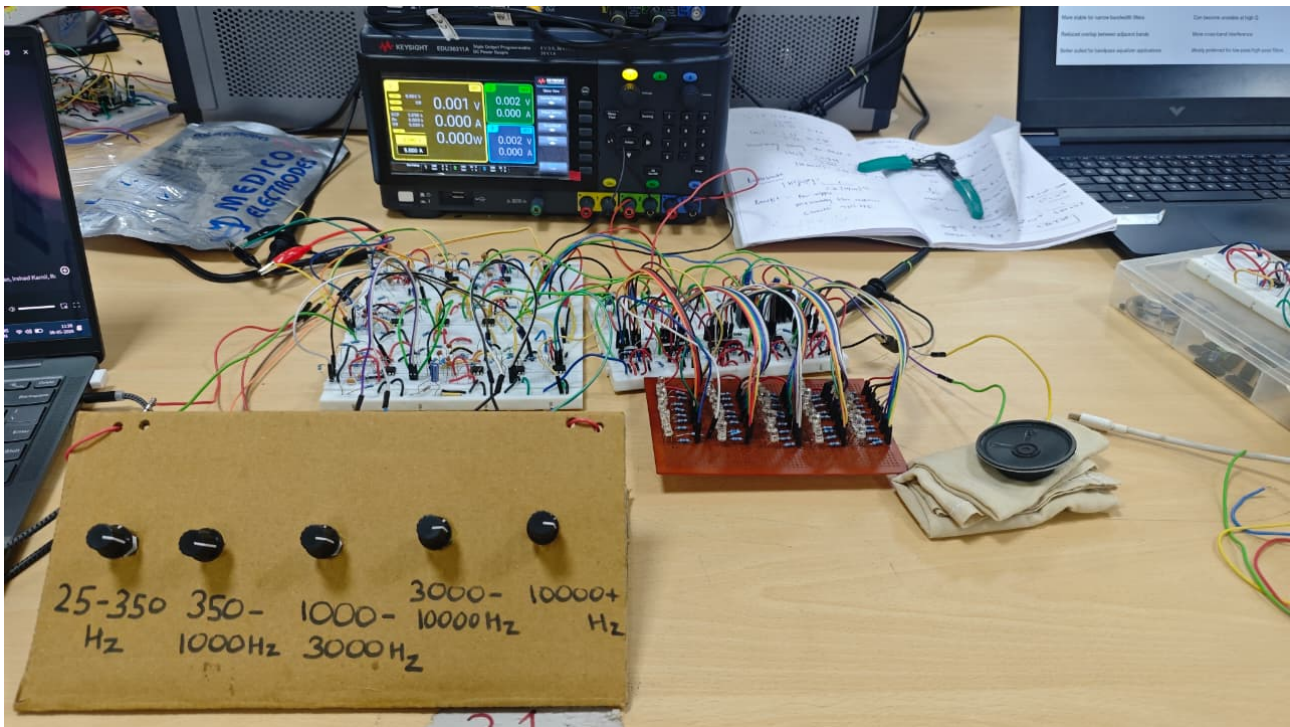


Figure 16: Final Circuit

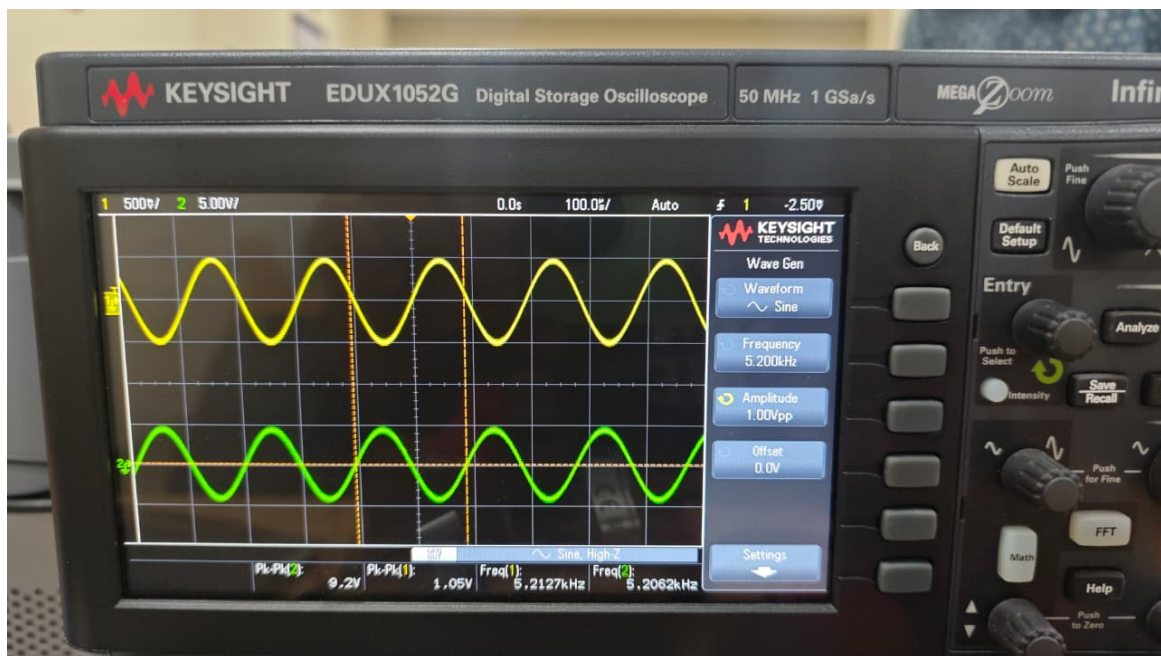


Figure 17: Output with input as 1 Vpp and all gain stages set to max values

5.2.1 Bode Plots:



Figure 18: Filter for 25-350 Hz filter



Figure 19: Filter for 350 Hz - 1 kHz Hz filter



Figure 20: Filter for 1 kHz - 3 kHz filter



Figure 21: Filter for 3 kHz - 10 kHz filter

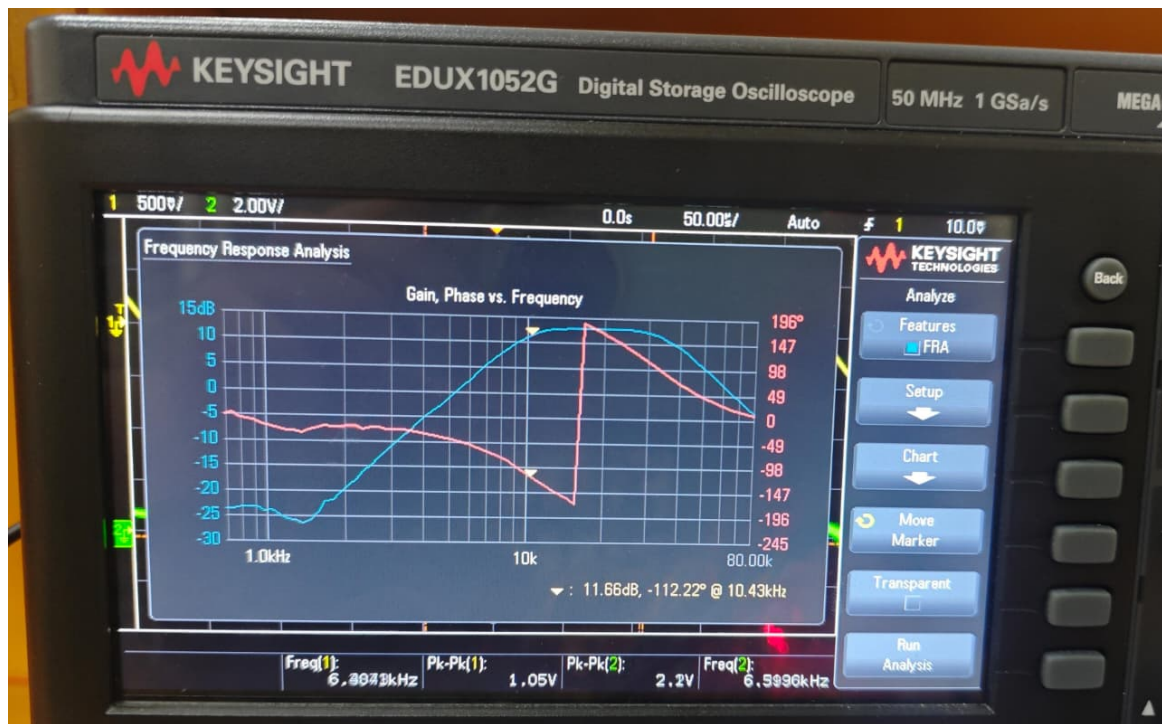


Figure 22: Filter for 10 kHz - 20 kHz filter

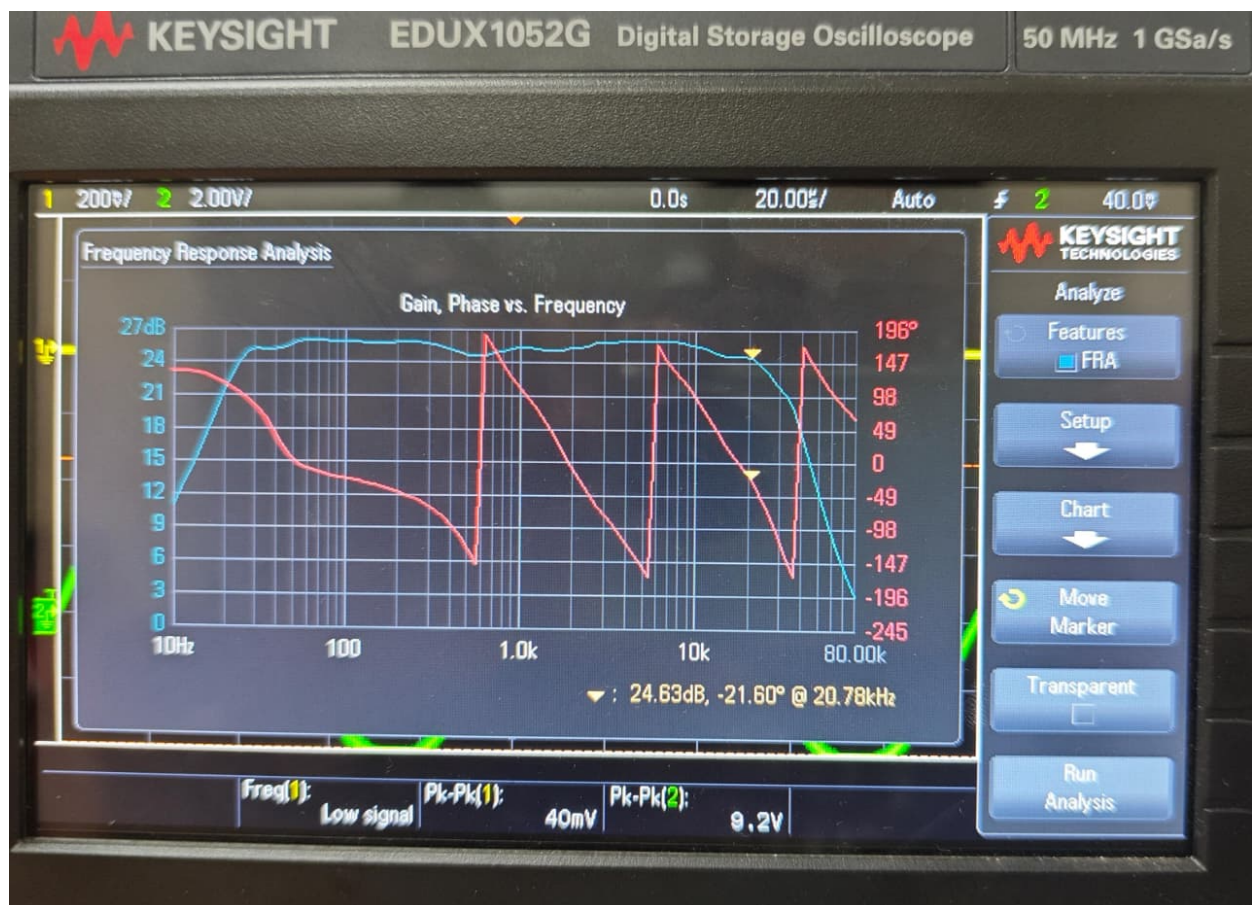


Figure 23: Final Result after adding all bands

5.2.2 Inputs

Here is an example of input received from the aux cable from the laptop. The volume of laptop was set accordingly to set the amplitude of input to 1 Vpp.



Figure 24: Sound Input

5.2.3 Analyzer

Here is the output of the spectrum analyzer for an input frequency of 5 kHz. Due to non-ideal roll-offs we can see a few LEDs of the other bands glowing. To perfectly understand the role of the analyzer it is advised to watch the demonstration videos attached at the last page.

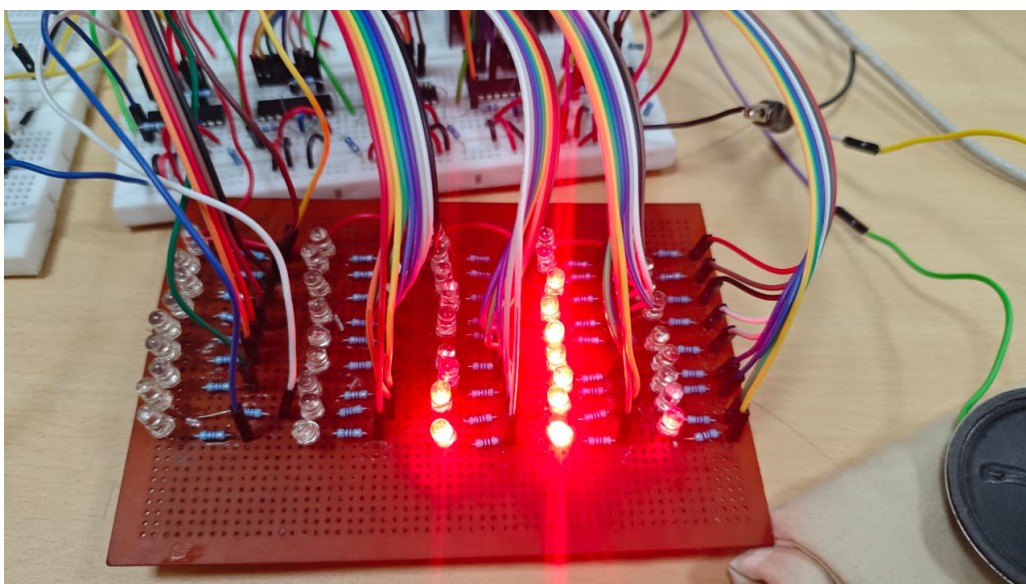


Figure 25: Analyzer output for input frequency = 5kHz

6 Measurements and Experimental Observations

The input signal applied to the circuit was assumed to have an amplitude of:

$$V_{in} = 1 \text{ Vpp} \quad (14)$$

The maximum achievable gain of each filter stage was set such that the maximum amplitude of the signal after passing through each filter stage was approximately:

$$V_{\max} = 8.6 \text{ Vpp} \quad (15)$$

This adjustment ensured nearly uniform amplification across the stages while also preventing excessive clipping and output distortion in the amplifier section. Reason for selecting this value was

After connecting the power amplifier stage, the output voltage of each filter stage was measured experimentally. It was observed that the output amplitude varied approximately within the range:

$$9 \text{ Vpp} \leq V_{out} \leq 13 \text{ Vpp} \quad (16)$$

depending on the specific filter stage and operating conditions.

The observed voltage amplification across the stages confirms that the power amplifier provided sufficient gain to the filtered signal while maintaining stable operation across different frequency bands.

7 Analysis and Discussion

While the system successfully demonstrated 5-band audio equalization, several discrepancies between theoretical models and hardware implementation were observed, presenting specific design challenges during testing.

7.1 Cross-Band Interference

Despite using a maximally flat Butterworth response, overlap between adjacent frequency bands caused unwanted mixing of frequencies. This spectral overlap reduced the overall selectivity of the equalizer, meaning large amplitude adjustments to one frequency band slightly colored the frequencies of its neighboring bands.

7.2 Complex Fourth Order Bandpass Filter Design

Designing and implementing the fourth-order MFB Butterworth bandpass filters was challenging due to the strict requirements for precise cutoff frequency tuning and gain adjustment. Maintaining a stable filter response across the cascaded operational amplifier stages required exact impedance matching and careful selection of feedback components.

7.3 Phase Response Mismatch

Phase shifts introduced by the higher-order active filters affected signal synchronization. Because each band undergoes varying degrees of phase shift (especially near its cutoff frequencies), the recombined audio signal at the summing amplifier experienced minor phase cancellations, which slightly impacted the overall audio quality and coherence.

7.4 Maintaining Audio Quality

Achieving proper amplification without introducing distortion or noise required careful circuit optimization. A notable design limitation is the operational amplifier's Gain-Bandwidth Product (GBWP) and slew rate. If an op-amp with insufficient GBWP is used for the 10 kHz - 20 kHz band, it struggles to maintain the required gain, distorting the response. Furthermore, the Class AB amplifier required precise diode biasing to prevent trace amounts of crossover distortion at low amplitudes.

7.5 Parasitic Effects

Breadboard implementation introduces significant parasitic capacitance between adjacent rows (typically 2 – 5 pF) and parasitic inductance in jumper wires. At lower frequencies (Bands 1 and 2), these effects are negligible. However, in Band 4 and Band 5 (up to 20 kHz), parasitic capacitance can inadvertently form low-pass filters, leading to high-frequency roll-off and reduced amplitude response.

7.6 Suggestions for Improvement

Grounded in the technical observations above, the following improvements are suggested for future iterations of this project:

1. **PCB Implementation:** Migrating the circuit from a breadboard to a properly routed Printed Circuit Board (PCB) with a dedicated ground plane would drastically minimize parasitic capacitance, improve stability, and reduce the overall noise floor.

2. **Increasing the Number of Bands:** Expanding the system from 5 bands to a 7-band or 9-band graphic equalizer configuration would provide much finer, granular control over the audio spectrum, allowing users to target specific resonant frequencies more accurately.
3. **6th Order Feedback System:** Upgrading the active filters from a 4th-order to a 6th-order MFB system would steepen the roll-off characteristics outside the passband. This would directly solve the cross-band interference issue by drastically improving band selectivity.

8 Conclusion

The design, simulation, and implementation of a 5-band audio equalizer were successfully executed. By utilizing Multiple-Feedback Butterworth bandpass filters, the audio spectrum was accurately divided into five distinct controllable bands ranging from 20 Hz to 20 kHz. The integration of envelope detectors, a summing amplifier, an LED audio analyzer, and a Class AB power amplifier resulted in a fully functional, user-adjustable audio processing system. Technical analysis confirmed that while component tolerances, phase shifts, and parasitic effects introduce minor deviations, the fundamental principles of active analog filtering were effectively demonstrated and validated against theoretical expectations.

Project Demonstration & Resources:

A complete video demonstration showcasing the fully functional 5-band audio equalizer and real-time LED spectrum analyzer can be accessed via the following link:

<https://drive.google.com/drive/folders/1HjKUTuw6CYNPMXkxBrCi83NWthcEkepB>

This also includes the filter stage calculations.

References

- [1] A. Sedra and K. Smith, *Microelectronic Circuits*, 8th ed. Oxford University Press, 2019.
- [2] Texas Instruments, *Specific Op-Amp Datasheet (e.g., OPA4187)*, 2021.
- [3] <https://www.nutsvolts.com/magazine/article/led-graph-circuitsputinreferences>.
- [4] https://www.electronics-tutorials.ws/filter/filter_7.html
- [5] <https://spiceman.net/adder-circuit/>